



Loreley Selene Lago

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PROFILE

Physical oceanographer with a PhD from the public University of Buenos Aires, specializing in ocean circulation and forcing mechanisms on continental shelves. Doctoral research focused on the Southwestern Atlantic Continental Shelf using in situ ADCP measurements, hydrographic observations, satellite altimetry, and reanalysis outputs. Experienced in oceanographic fieldwork, statistical analysis, and advanced programming. Currently a postdoctoral investigator at the Woods Hole Oceanographic Institution, studying Pacific Arctic circulation and water mass variability and their links to climate-driven changes in marine ecosystems, including the increasing occurrence of harmful algal blooms.

EDUCATION

- 2017 – 2022 **Atmospheric and Ocean Research Institute, Faculty of Exact and Natural Sciences, University of Buenos Aires**
PhD studies in Atmosphere and Ocean Science

Thesis: Variability of the ocean circulation in the Patagonia continental shelf from in-situ and satellite data, and its possible forcing
Dir: Dr. Martín Saraceno
Co-Dir: Mg. Patricia Martos
- 2008 - 2016 **Atmospheric and Ocean Research Institute, Faculty of Exact and Natural Sciences, University of Buenos Aires**
Degree in Physical Oceanography (equivalent to a B.Sc. and M.Sc. in science)

Thesis: Validation of satellite altimetry data in the San Matias Gulf, at the Southwestern Atlantic Ocean, around 41.2°S and 2 km from the coast
Dir: Dr. Martín Saraceno
Co-Dir: Dra. Laura A Ruiz-Etcheverry

ACADEMIC BACKGROUND

Published Articles in Peer-Reviewed International Journals

Lago, L.S., Pickart, R., Lin, P., Bahr, F., Fachon, E., Brosnahan, M., Pathare, M., Muhlbach, E., Horn, K., Rajagopalan, A., Anderson, D. (2025). Physical drivers of a massive harmful algal bloom in the Chukchi Sea in summer 2022. *Journal of Geophysical Research: Oceans*, 130(4),

<https://doi.org/10.1029/2024JC021624>

Lago, L.S., Saraceno, M., Piola, Ruiz-Etcheverry, L. A. (2021). Volume transport variability on the northern Argentine Continental Shelf from in situ and satellite altimetry data. *Journal of Geophysical Research: Oceans*, 126(2), e2020JC016813. <https://doi.org/10.1029/2020JC016813>

Lago, L. S., Saraceno, M., Martos, P., Guerrero, R. A., Piola, A. R., Painiagua, G. F., Ferrari, R., Artana, C. I., Provost, C. (2019). On the wind contribution to the variability of ocean currents over wide continental shelves: a case study on the northern Argentine continental shelf. *Journal of Geophysical Research: Oceans*, <https://doi.org/10.1029/2019JC015105>

Lago, L. S., Saraceno, M., Ruiz-Etcheverry, L. A., Passaro, M., Oreiro, F., D'Onofrio, E. E., González, R. (2017). Improved Sea Surface Height from Satellite Altimetry in Coastal Zones: A Case Study in Southern Patagonia. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, <https://doi.org/10.1109/JSTARS.2017.2694325>

Pickart, R., Spall, M., Bahr, F., **Lago, L.S.**, Lin, P., Pacini, A., Huang, J., Arrigo, K., Mills, M., van Dijken, G., McRaven, L., Roberts, S. (2025). Vertical carbon export during a phytoplankton bloom in the Chukchi Sea. Part 1: Physical setting and frontal subduction. *Journal of Geophysical Research: Oceans*, 129(11), e2024JC021465. <https://doi.org/10.1029/2024JC021465>

Zhang, X., Lin, P., Pickart, R.S., Torres, D., Bahr, F., Houk, A., **Lago, L.S.**, Yang, X. Seasonal to interannual variation of the Beaufort Shelfbreak Jet using two decades of mooring data, 2002-2022. *Journal of Geophysical Research*, 130, e2025JC022555. <https://doi.org/10.1029/2025JC022555>

Lefebvre, K. A., Campbell, C. M., Divine, L. M., Melovidov, P., Hellen, H., Huntington, K. B., Bowers, E. K., Rouse, N., Fachon, E., Farrugia, T. J., Pickart, R. S., Lin, P., **Lago, L. S.**, Bahr, F., Furst, E., Duncan, C. G., Charapata, P., Anderson, D. M. & Kurtay, G. (2025). Saxitoxin Linked to Deaths of Northern Fur Seals in the Southeast Bering Sea. *Marine Mammal Science*, e70028 <https://doi.org/10.1111/mms.70028>

Fachon, E., Pickart, R. S., Sheffield, G., Pate, E., Pathare, M., Brosnahan, M. L., Muhlbach, E., Horn, K., Spada, N., Rajagopalan, A., Lin, P., McRaven, L., **Lago, L.S.**, Huang, J., Bahr, F., Stockwell, D., Hubbard, K., Farrugia, T., Lefebvre, K., Anderson, D. M. (2025). Tracking a large-scale and highly toxic Arctic algal bloom: Rapid detection and risk communication. *Limnology and Oceanography Letters*, 10(1), 62-72. <https://doi.org/10.1002/lol2.10421>

Fachon, E., Schrage, K., **Lago, L.S.**, Pathare, M., Lim, S., Goethel, C. Baby Bivalves (just submitted) A high-density multi-species bivalve spawning event in the southern Chukchi Sea. *Marine Ecology Progress Series*, under review.

Lim, S.M., Mills, M.M., Payne, C.M., Fachon, E., Anderson, D.M., Pickart, R.S., **Lago, L.S.**, and Arrigo, K.R. Urea dominates nitrogen uptake in the Pacific Arctic Ocean during a summer bloom of *Alexandrium catenella*. *Limnology and Oceanography*, under review.

Schrage, K.R., Weston, J.N.J., Kraberg, A., McPherson, R., Von Appen, W.J., **Lago, L.S.**, S. Meyer-Kaiser, K.S. Diversity, drivers, and dispersal of high Arctic meroplanktonic communities. Manuscript in prep.

Abdalla, S., Kolahchi, A. A., Ablain, M., Adusumilli, S., Bhowmick, S. A., Alou-Font, E., ... & Hamon, M. (2021). Altimetry for the future: Building on 25 years of progress. *Advances in Space Research*. <https://doi.org/10.1016/j.asr.2021.01.022>

Paniagua, G. F., Saraceno, M., Piola, A. R., Guerrero, R. A., Provost, C., Ferrari, R., **Lago, L. S.**, Artana, C. I. (2018). Malvinas Current at 40°-41° S: First Assessment of Temperature and Salinity Temporal Variability. *Journal of Geophysical Research: Oceans*, <https://doi.org/10.1029/2017JC013666>

Dias, A. C. E., Paleček, D., Pardo, J., Hartman, S., **Lago, L.**, Rech, T., & Almeida, L. L. Useful tools for Environmental Education: Spreading knowledge in innovative and engaging ways. CHALLENGES IN OCEAN GOVERNANCE IN THE VIEWS OF EARLY CAREER SCIENTISTS, 208.
<https://doi.org/10.11606/9786587773384>

Technical Reports

Lago, L. S., Saraceno, M., Martos, P., Guerrero, R. A., Piola, A. R., Painiagua, G. F., Ferrari, R., Artana, C. I., Provost, C. (2021). Corrientes oceánicas en la plataforma continental Bonaerense. Technical report, Instituto de Investigación y Desarrollo Pesquero (INIDEP).

Presentations in Congresses and Scientific Reunions. WITH ATTENDANCE

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| 2025 | <p>Polar Science Gordon Research Conference and Seminar</p> <p>Lucca, Italy</p> <p>Poster: The role of physical properties and circulation in shaping a massive harmful algal bloom in the Pacific Arctic during summer 2022.</p> |
| 2024 | <p>30 Years of progress in radar altimetry symposium</p> <p>Montpellier, France</p> <p>Poster: Water mass distribution and circulation in the Chukchi Sea in relation to a massive harmful algal bloom during summer 2022.</p> <p>Arctic Sub-Arctic Ocean Fluxes (ASOF)</p> <p>Halifax, Canada</p> <p>Oral: Water mass distribution and circulation in the Chukchi Sea in relation to a massive harmful algal bloom during summer 2022.</p> <p>Ocean Science Meeting (OSM)</p> <p>New Orleans, USA</p> <p>Oral: Water mass distribution and circulation in the Chukchi Sea in relation to a massive harmful algal bloom during summer 2022</p> |
| 2020 | <p>Ocean Surface Topography Science Team (OSTST)</p> <p>Virtual</p> <p>Poster: Analysis of a 25-year long volume transport time series derived from satellite altimetry data and in situ measurements.</p> |
| 2019 | <p>Ocean Surface Topography Science Team (OSTST)</p> <p>Chicago, USA</p> <p>Oral: Volume Transport from In-situ and Altimetry Data Over a Wide Continental Shelf.</p> <p>Oral: Volume transport and modes of variations of the Malvinas Current at 44.7°S from satellite altimetry and current-meter velocities.</p> |
| 2018 | <p>X National Conference on Marine Science (JNCM)</p> <p>Buenos Aires, Argentina</p> <p>Oral: Corrientes oceánicas in-situ y satelitales en un área de alta productividad biológica de la plataforma continental Argentina.</p> |

- 2018 **11th Coastal Altimetry Workshop**
 Frascati, Italia
 Oral: The performance of satellite altimetry currents in a wide continental shelf
- 2017 **10th Coastal Altimetry Workshop**
 Firenze, Italia
 Poster: Improved sea surface height from satellite altimetry in coastal zones: a case study in southern Patagonia
 Oral: Satellite altimetry in the continental shelf of the southwestern Atlantic, Argentina
- 2016 **Ocean Surface Topography Science Team (OSTST)**
 La Rochelle, France
 Poster: CASSIS-Malvinas: Southwestern Atlantic currents from in-situ and satellite altimetry data project

VISITS AND EXCHANGES

- 2025 **Woods Hole Oceanographic Institution**
 Woods Hole, USA
 Invited speaker for Polar Lunch seminar series
 Subject: Where do they come from? Where do they go? A collaborative investigation of harmful algal blooms in the Pacific Arctic.
- University of Massachusetts Dartmouth**
 New Bedford, USA
 Seminar invited speaker at DEOS S Mast
 Subject: The role of physical drivers in the initiation, growth and distribution of harmful algal blooms in the Pacific Arctic.
- Woods Hole Center for Oceans and Human Health Meeting**
 Woods Hole, USA
 Invited talk: Physical Drivers of a Massive Harmful Algal Bloom in the Northern Bering and Chukchi Seas in Summer 2022.
- 2024 **University of Exeter**
 Department of Earth and Environmental Sciences
 Cornwall, UK
 Invited talk: Continental Shelf Dynamics: How local winds modulate the circulation in the Southwestern Atlantic Ocean.
- CNR, ISTITUTO DI SCIENZE MARINE (ISMAR)**
 La Spezia, Italy
 Invited talk: The role of physical properties and circulation in shaping a massive harmful algal bloom in the Pacific Arctic during summer 2022.
- University of Milano - Bicocca**
 Department of Earth and Environmental Sciences
 Milan, Italy

Invited talk: Continental Shelf Dynamics: How local winds modulate the circulation in the Southwestern Atlantic Ocean.

2016 **One-month internship within CASSIS project**

Advisor: Christine Provost
L'OCEAN, Sorbonne Université, Paris, France

TEACHING BACKGROUND

2021 - 2022 **Head teaching assistant**

Physical Oceanography, Department of Atmospheric and Ocean Sciences, School of Hard and Natural Sciences, University of Buenos Aires
Introduction to Atmospheric and Ocean Sciences. Introduction to Coastal Engineering. Satellite Oceanography

2018 **Teaching assistant**

Physical Oceanography, Department of Marine Science School of Hard and Natural Sciences, National University of Mar del Plata
Physical Oceanography (1st period)

2020 **Assistant of a degree thesis in physical oceanography**

Student: Dinora Noelia García Santa Cruz
Director: Dr. Martín Saraceno
Title: Study of the tidal height and currents from in-situ data and models in the Argentine Continental Shelf. Dissertation: 02/10/2020
School of Hard and Natural Sciences, University of Buenos Aires

AWARDED GRANTS

- 2025 **Co-PI for NSF-OPP Polar Research Operations**
Woods Hole Oceanographic Institute (WHOI), MA, United States
submitted in July 2025
Subject: Harmful algal bloom dynamics, drivers, and impacts in the Eastern Bering Sea.
- 2023 – present **Postdoc grant**
Woods Hole Oceanographic Institute (WHOI), MA, United States
Subject: Water mass distribution and circulation of the Pacific Arctic, Chukchi Sea, and its connection to the massive harmful algal bloom detected in summer 2022.
Dir: Dr. Robert Pickart
Co-Di: Dr. Peigen Lin
- 2022 - 2023 **Postdoc temporary grant (6 months)**
National Council of scientific and technical research (CONICET), Argentina
Subject: Study of the intrusions of the Malvinas Current into the Argentine continental Shelf, along the shelf break in the Southwestern Atlantic Ocean by using in-situ, satellite and reanalysis data.
Dir: Dr. Martín Saraceno
Co-Di: Prof. Alberto Piola
- 2017 - 2022 **PhD grant in strategic subjects**
National Council of scientific and technical research (CONICET), Argentina
- 2015 - 2016 **Grant for the formation of human resources in Oceanography (master)**
Scientific Technological Articulation Secretariat of the Ministry of Science, Technology and Productive Innovation, Argentina

LANGUAGES

- SPANISH** Native
- ENGLISH** Advanced Level (International Certificate)
First Certificate in English, A grade, 2006, University of Cambridge – ESOL Examinations
- FRENCH** Advanced Level (International Certificate)
DEL F B2, 2011, Ministère de l'éducation nationale, de la jeunesse et de la vie associative
- ITALIAN** Advanced Level
Language University Center (CUI) of the University of Buenos Aires
- CHINESE** Initial Level
Language University Center (CUI) of the University of Buenos Aires

OTHER SKILLS

Advanced proficiency in MATLAB, Python, and CDO for statistical analysis and scientific data visualization. Peer reviewer for international journals, including JGR: Oceans, Ocean Dynamics, and Deep-Sea Research.

Participation in oceanographic cruises includes the Upstream Pathways of the Faroe Overflow (UFO) cruise aboard R/V Revelle (August–September 2025); AU04/2017 Agujero Azul cruise aboard ARA Austral (September 2017, Southwestern Atlantic); CASSIS cruises aboard ARA Puerto Deseado (July 2016, May 2017) and coastguard vessel Tango SB-15 (November 2015), Southwestern Atlantic Ocean.

SCIENCE COMMUNICATION AND MANAGEMENT ACTIVITIES

2024 **Elected postdoctoral representative**

Woods Hole Oceanographic Institution Postdoctoral Association (PDA)

2024 **Physical Oceanography postdoc lunch organizer**

Organized weekly postdoc lunches at Woods Hole Oceanographic Institution, facilitating discussions with Department staff on career paths and institutional opportunities.

2024 **Science Communication (SciComm) workshop**

PASS – People’s Awards for Scientific Storytelling, Short, elevator-pitch like presentations of complex research with peer feedback, ~50 participants (WHOI postdocs, undergraduate and graduate students, scientific and technical staff), Woods Hole Oceanographic Institution.

2021 - 2023 **La Oceanoteca**

Co-founded and led an independent Spanish-language science communication team in Physical Oceanography, developing podcast content to make ocean science accessible to the general public and to promote appreciation and understanding of ocean processes.

Members: Dr. Daniela Belén Risaro and Dr. Loreley Selene Lago

IG: @laoceanoteca

Spotify: @laoceanotec

2014 – 2015 **Physical Oceanography Science Communicator**

Faculty of Exact and Natural Sciences, University of Buenos Aires — Department of Science Popularization (SECC-B). Delivered science communication and outreach activities to promote awareness of high-impact but underrepresented scientific careers in Argentina, as Physical Oceanography. Activities included hands-on demonstrations of scientific experiments and outreach talks on scientific concepts and career paths at secondary schools and public events.